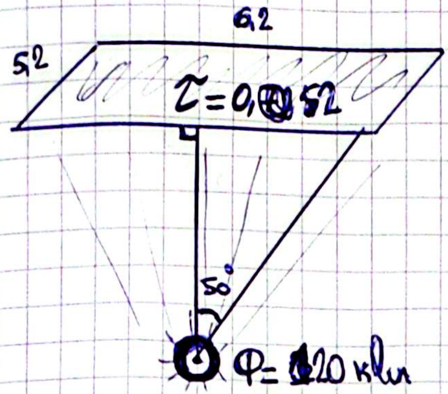
Homework 6 | Mujabbar Badalov | MPA-SVT.

**Problem 11:** A ceiling with dimensions 5,2 x 6,2 and its transmittance 0,52. Illuminated by 120 klm. Illuminance E=? , $L$ in vertical and in the direction 50° from vertical.




$S = 5{,}2 \times 6{,}2 = 32{,}24$

$E = \dfrac{\Phi}{S} = \dfrac{120 \cdot 10^3}{32{,}24} = 3722{,}09$ lux

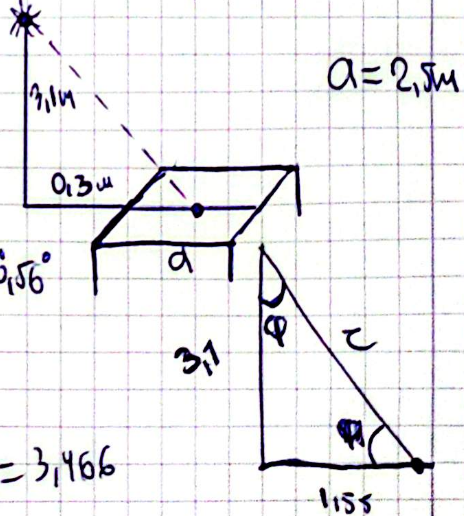
Luminance of suspended ceiling $\left[ L_\tau = \dfrac{3722{,}09 \cdot 0{,}52}{\pi} = \boxed{616{,}08 \ cd \cdot m^{-2}} \right.$

Luminance in vertical direction $I_0 = L_\tau \cdot S = 616{,}08 \cdot 32{,}24 = 19862{,}57$ cd

Luminance in angle direction $I_{50°} = I_0 \cdot \cos 50° =$ cd 12767,41 cd

**Problem 12**

Specific power 26 lm/W  P=?
in the centre illuminance must be 90 lx




So the angle will be: arctan $\dfrac{0{,}3 + 2{,}5/2}{3{,}1} = 26{,}56°$

$E = \dfrac{\Phi \cos\varphi}{4\pi r^2}$ , $\Phi = P \cdot M$

$r = 3{,}466$

$P_{req} = \dfrac{90 \cdot 4\pi \cdot (3{,}466)^3}{26 \cdot 3{,}1} \approx 584{,}2$ W

$P_{actual} = 600$ W

Then $E_{actual} = \dfrac{600 \cdot 26 \cdot 3{,}1}{4\pi \cdot (3{,}466)^3} \approx 92{,}43$ lx